22 (a) Write $3x^2 - 24x + 38$ in the form $a(x - b)^2 - c$ where a, b and c are integers.

$3x^2 - 24x + 38 = 3(x^2 - 8x) + 38$

1 mark
Factor out 3.

$= 3[(x - 4)^2 - 16] + 38$

1 mark
Complete the square.

$= 3(x - 4)^2 - 48 + 38$

$= 3(x - 4)^2 - 10$

1 mark
a = 3, b = 4, c = 10.

.....

(3)

(b) Hence, or otherwise, write down the coordinates of the turning point on the graph of $y = 3x^2 - 24x + 38$

Turning point = (4, -10)

1 mark
Minimum when x = 4.

(....., ...)

(1)

(Total for Question 22 is 4 marks)